

Sunipun Talukder

AI Engineer

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SUMMARY

AI Engineer with a mathematics background, building generative AI and computer vision systems that ship to millions of users. Four years across diffusion models, image enhancement and on-device inference.

EXPERIENCE

AI Engineer December 2022 - Present

BrainCraft Ltd — Dhaka, Bangladesh

- Engineered the generative AI image pipeline behind Cartoon.ai, propelling it to top-grossing status on the App Store
- Built AI photo enhancement modules — upscaling, deblurring, denoising and artifact removal — improving image quality for over 500K users
- Co-developed and deployed a semantic image segmentation model serving a user base exceeding 10 million
- Implemented text-to-image and text-to-GIF generation on stable diffusion models in Python, combining modern ML techniques with classical image processing

Software Engineer January 2022 - November 2022

BrainCraft Ltd — Dhaka, Bangladesh

- Designed an image-to-video module on OpenGL ES and the MediaCodec API for the GIFMaker and Add Music to Video Android apps, together passing 250K downloads
- Wrote a core video encoder/decoder in C++ on the Android NDK, raising efficiency on low-end devices and expanding device availability by 22%
- Led a team of four delivering the BgRemover and Slideshow apps using MVVM, custom rendering and advanced video encode/decode

Software Engineer January 2021 - December 2021

LiiLab — Sylhet, Bangladesh

- Architected the Intro Maker app on Canvas and native Android APIs, reaching over 100K Play Store downloads
- Rebuilt GkQuiz's rank list on Firebase Realtime Database and Firestore for a 4x performance gain, serving over 250K users

SKILLS

AI & Machine Learning	Deep Learning · Diffusion Models · Computer Vision · Image Segmentation · Super Resolution · LLMs & RAG · On-Device AI · Model Optimization
ML & Data Tooling	PyTorch · Transformers · OpenCV · NumPy · Scikit-learn · FastAPI · Docker · Gradio
Languages	Python · C++ · Kotlin · Java · Swift
Mobile & Graphics	Android (Kotlin/Java) · Android NDK · OpenGL ES · GLSL Shaders · Metal · MediaCodec · iOS (Swift)
Data & Infrastructure	Vector Databases · MySQL · SQLite · Git · Agile · MVVM & MVC
Algorithms	Data Structures & Algorithms · Dynamic Programming · Graph Theory · Number Theory

EDUCATION

Shahjalal University of Science and Technology 2015 - 2020

Bachelor's in Mathematics

COMMUNITY & MENTORING

Admin & Community Leader Present

Shuddho Social Org

- Leading community development initiatives and fostering social responsibility
- Organizing events and managing volunteer teams for social impact projects

DS & Algorithms Mentor 2019 - Present

Independent

- Mentored 200+ students in data structures and algorithms
- Guided students through competitive programming and technical interviews

SELECTED CERTIFICATIONS

Machine Learning in Production — DeepLearning.AI (Coursera), July 2024

Introduction to Generative AI — Google Cloud (Coursera), January 2024

Machine Learning — Stanford University (Coursera), December 2020

On-Device AI — DeepLearning.AI & Qualcomm, 2024